

Aryan Jain

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Education

- 2024–2027 **BEng Electronic and Information Engineering, Imperial College London**, London, UK
Relevant Coursework: Programming for Engineers, Digital Electronics & Computer Architecture, Mathematics for Engineers, Circuits, Software Engineering.
- 2018–2022 **International Baccalaureate (41/42)**, United World College of South East Asia – East Campus, Singapore

Experience

- Jul 2025 – **Software Development Intern, Accenture**, Singapore
- Sep 2025
- Implemented backend workflow automation in C#, integrating validation layers and data preprocessing pipelines.
 - Built a cross-platform data slicer adopted by 60+ employees, reducing daily manual processing time by 40%.
 - Developed a microservice message-flow visualisation tool using Mermaid diagrams, enabling 100+ developers to trace system-level interactions.
 - Collaborated with QA teams to resolve SIT/UAT blocking issues, improving release stability for the project.
- May 2025 – **Trading Intern, Optiver – Imperial Trading Academy**, London, UK
- Jun 2025
- Completed intensive training in options, derivatives pricing, market-making, and execution algorithms.
 - Ranked in the top tier for decision quality and PnL stability across competitive trading simulations.
- Sep 2022 – **Platoon Sergeant, Guards Formation, Singapore Armed Forces**, Singapore
- Jul 2024
- Led and trained a 60-person platoon in advanced infantry tactics, weapons proficiency, and physical conditioning.
 - Coordinated mission planning and safety oversight for high-intensity field exercises.
 - Awarded Commander's Coin and Certificate of Recognition for leadership excellence.

Projects

- C, POSIX **Multithreaded UDP Chat System**
- Threads, UDP Sockets
- Developed a multi-client chat system using a UDP client-server model with pthread-based concurrency.
 - Implemented sender and listener threads per client; server spawns request-servicing threads per packet.
 - Built a thread-safe shared client registry (reader-writer locks) supporting broadcast, private messaging, mute/unmute, rename, and admin kick.
 - Added a 15-message history buffer and an inactivity-removal engine using a min-heap with ping/pong logic.
- SystemVerilog, Verilator **5-Stage Pipelined RISC-V CPU + Write-Back Data Cache**
- Built an RV32I CPU with forwarding, load-use stalls, branch/jump flushing, and full pipeline registers.
 - Designed a 2-way set-associative write-back data cache with LRU replacement and multi-cycle miss handling.
 - Created custom Verilator testbenches and debugging harnesses; validated behaviour using a Gaussian histogram workload stressing hazards and cache coherence.
- C++, **Sensor-Integrated Rover**
- JavaScript, Embedded Sensors
- Engineered a rover featuring IR, ultrasonic, and radio sensors for signal detection, ranging, and classification.
 - Implemented sensor fusion logic for noise reduction and multi-sensor confirmation of environmental readings.
 - Designed closed-loop motor control for responsive navigation based on real-time sensor input.
 - Built a web-based dashboard using JavaScript to visualise live telemetry, system state, and signal traces.
- Hardware, Signal Processing **RF Communication System (CGCU Hackathon)**
- Co-designed an RF communication system using card-encoded LDR signals to encode/receive optical data.
 - Improved receiver sensitivity and signal encoding for robust short-range transmission.

Extracurricular Activities

- Vice President, Electrical Engineering Society**
Organised events for 200+ students, coordinated with industry partners, and secured sponsorships.
- Rocketry Society (Recovery Team)**
Developed a parachute deployment mechanism triggered at apogee using onboard sensors and microcontrollers.

Skills

- Programming:** Python, Java, C, C++, C#, SQL, SystemVerilog, Bash
- Tools & Technologies:** Verilator, Git, Visual Studio, Azure, Fusion 360, POSIX Threads, UDP Sockets